

Multiple Choice (10 marks)

1. Cholesterol and fatty acids absorbed from the intestinal lumen are packaged into which types of lipoproteins and secreted from the enterocytes into the lymph?
 - (a) Chylomicrons only
 - (b) LDL only
 - (c) VLDL and LDL
 - (d) Chylomicrons and HDL
 - (e) VLDL and HDL
2. A burial site where the body is allowed to decompose naturally without a casket is called a _____ cemetery
 - (a) Eco-friendly
 - (b) Sunshine
 - (c) Forest
 - (d) Rapid decay
 - (e) Organic
3. Older adults are able to improve their memories and reduce their anxiety about declining memory when
 - (a) They engage in social activities and interact with peers
 - (b) They practice mindfulness and meditation to reduce stress levels
 - (c) They engage in physical activity regularly
 - (d) They regularly solve puzzles and engage in brain-stimulating activities
 - (e) They learn that many aspects of memory do not decline and some even get better
4. The energy charge of the cell is:
 - (a) the extent to which the total adenine nucleotide pool is phosphorylated.
 - (b) the difference between the total energy input and output of the cell.
 - (c) generated by the sodium-potassium ATPase.
 - (d) the amount of ATP available in the cell.
 - (e) the energy required for the maintenance of the cell's structure.
5. The following amino acids are all essential components of the diet:
 - (a) leucine, methionine and threonine

- (b) leucine, cysteine and glycine
- (c) proline, tyrosine and serine
- (d) valine, glycine and arginine
- (e) methionine, serine and tyrosine

True / False (10 marks)

Answer True or False

- 6. True or False: Nearly one-third of grieving individuals appear to require some form of professional help.
- 7. True or False: A person weighing 62 kg with a drug dose of 15 mg/kg requires a total dose of 18.6 mg.
- 8. True or False: Type IIb muscle fibers are characterized by low fatigue resistance compared to other fiber types.
- 9. True or False: Meat and dairy products are consumed in lower quantities in Wales and Scotland compared to other regions of the UK.
- 10. True or False: A normal resting heart rate for an adult is considered to be between 70 and 90 beats per minute.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the relationship between optimism in older adults and longevity, analyzing how pessimistic outlooks may influence health outcomes and lifespan.
- 12. Discuss the energy expenditure of various organs in the human body, focusing on why the intestines have the lowest energy expenditure per unit of body mass.

End of Paper